Line Methods and Direction Set Methods in Multidimensional Optimization

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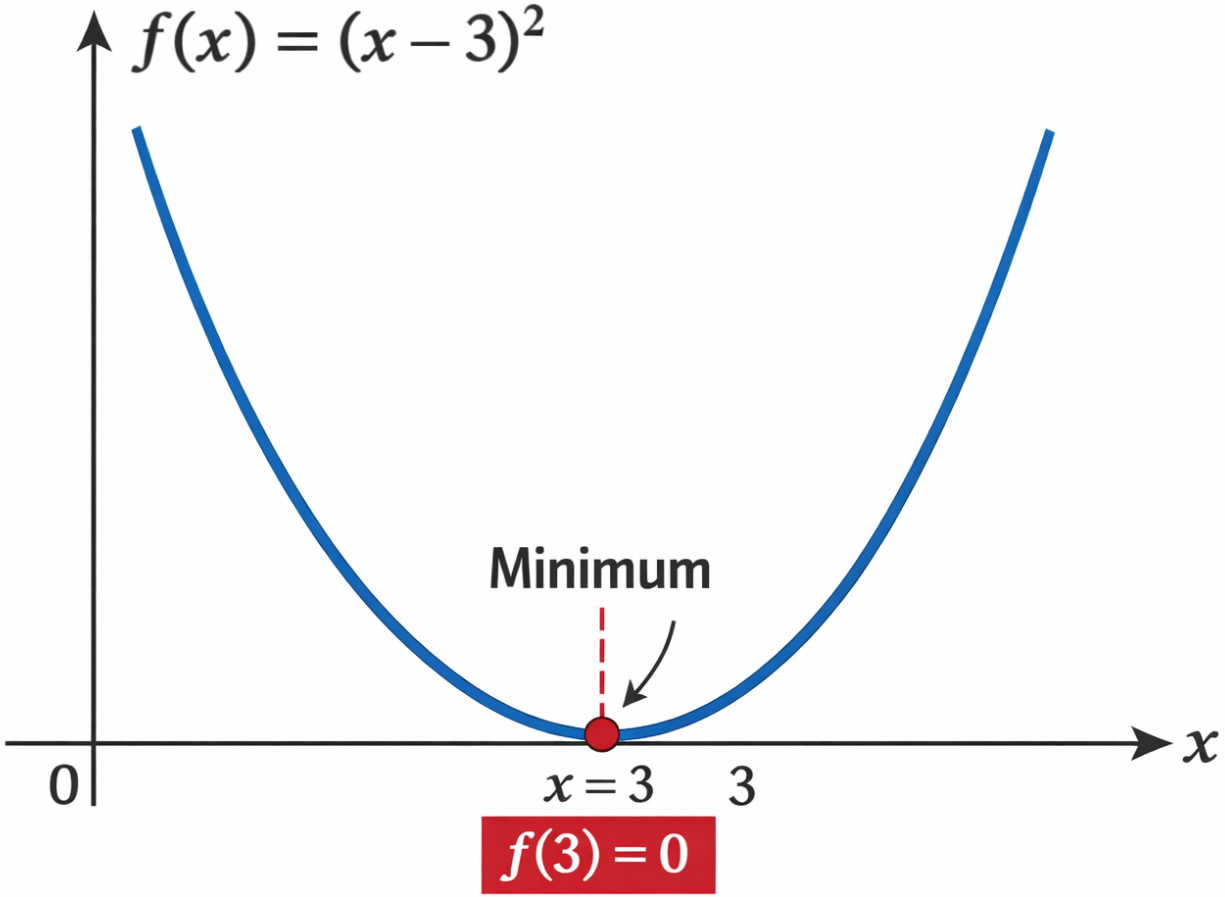
From 10.6 (linmin) to 10.7 (Powell's Method)

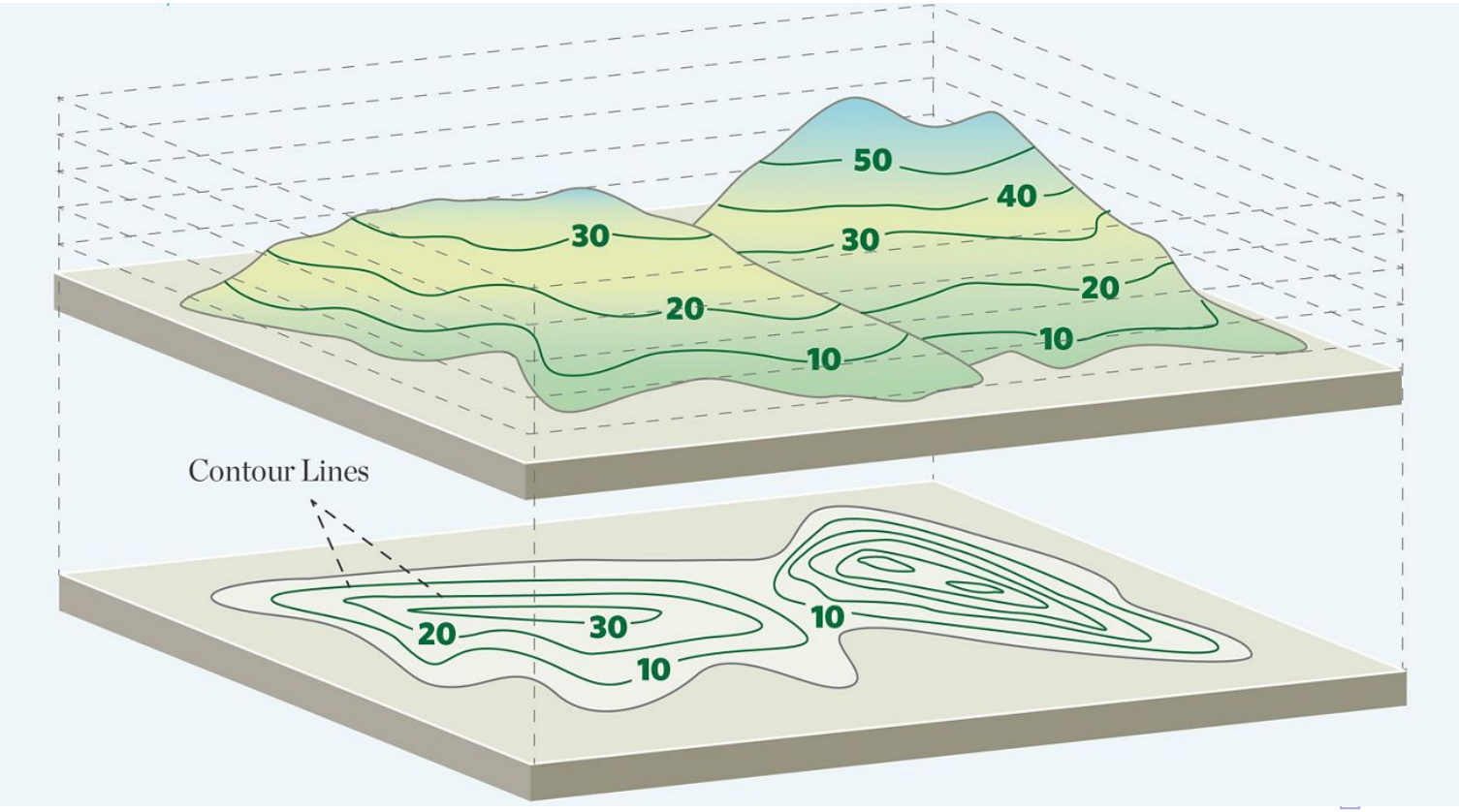
What Does It Mean to Minimize a Function?

- Goal:
- A function that maps inputs to outputs
- We want the input that produces the smallest output
- Example: f(x) = (x - 3)^2, minimum occurs at x = 3, f(3) = 0



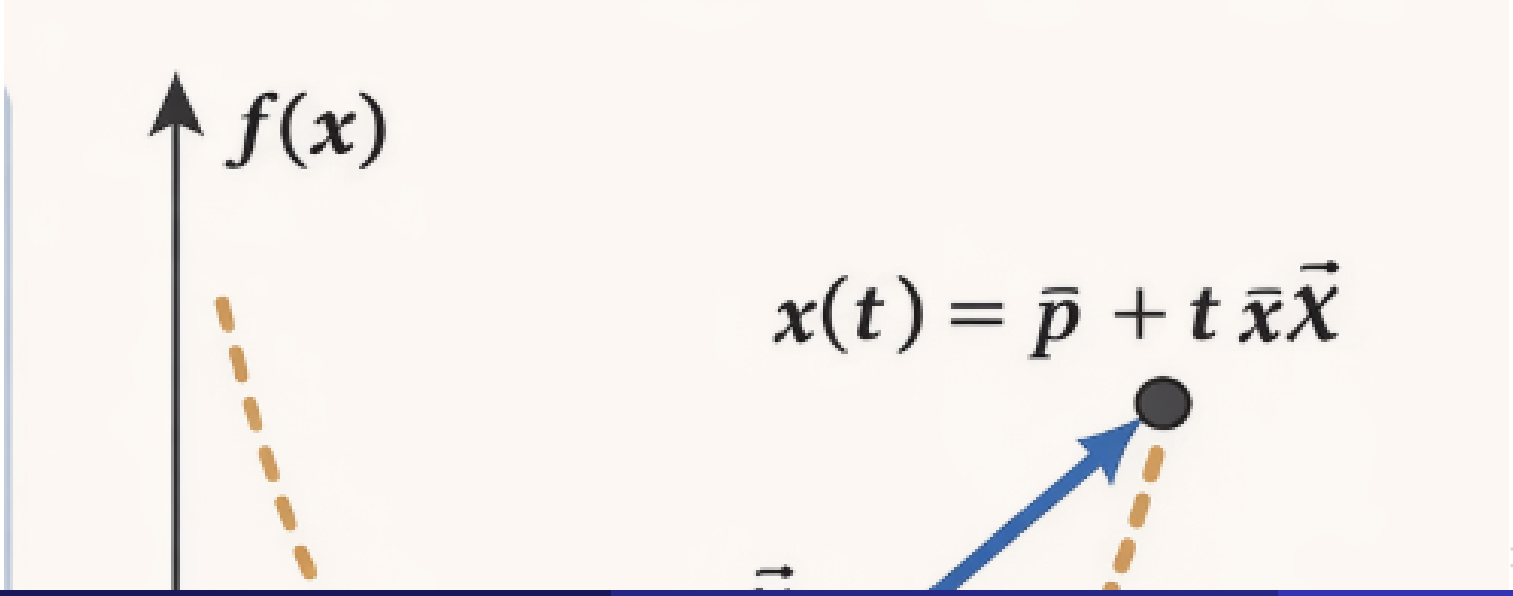
From One Variable to Many Variables

- One variable: f(x)
- Many variable: f(x,y), f(x,y,z), f(x)
- Represents a surface or landscape
- Hard to search directly
- Analogy: Finding the lowest point on a mountain



Key Idea of 10.6: Reduce to One Dimension

- Instead of searching everywhere:
- Pick a starting point
- Pick a direction
- Move along that straight line
- Mathematically:
- x(t)=p+t
- Define :
- g(t)=f(p+t)
- Thus the problem is now a one dimensional problem



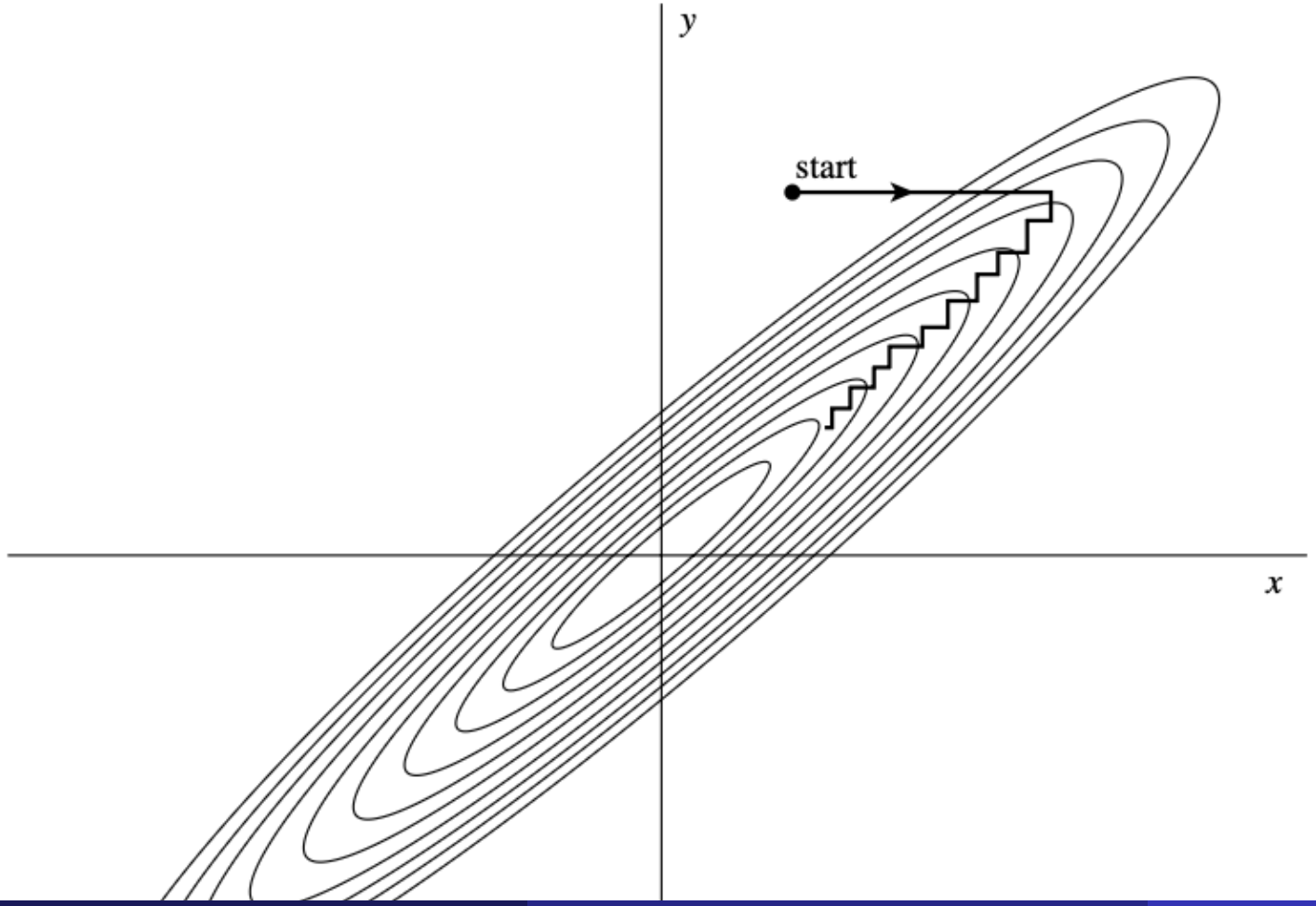
What linmin Does (10.6)

- Input:
- Current point, p
- Direction,
- function , f
- Output:
- Moves p to the min along that line
- Returns the function value

`linmin`: Given as input the vectors $\mathbf{P}$ and $\mathbf{n}$, and the function f, find the scalar λ that minimizes $f(\mathbf{P} + \lambda\mathbf{n})$. Replace $\mathbf{P}$ by $\mathbf{P} + \lambda\mathbf{n}$. Replace $\mathbf{n}$ by $\lambda\mathbf{n}$. Done.

Big Question: Which direction should we choose next

- From 10.6 we already know how to move to the best point along one chosen direction using linmin()
- If we choose directions poorly, the algorithm can be extremely slow.



Goal of Direction Set Methods (10.7)

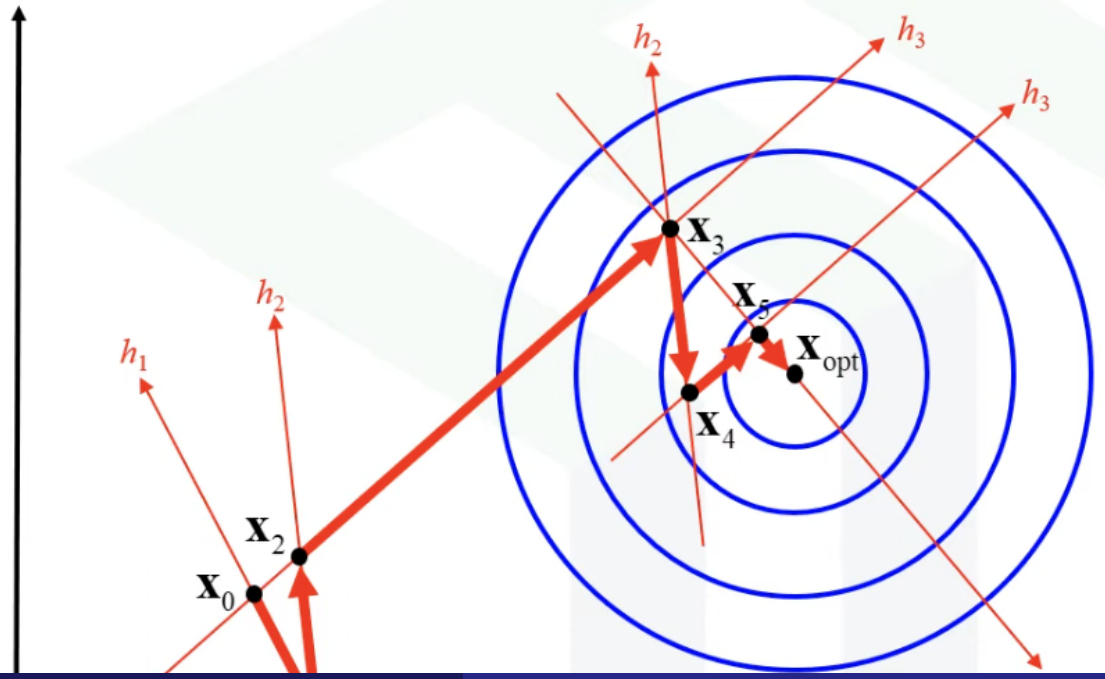
- Instead of fixed directions:
- Update directions as the algorithm runs.
- Learn directions that follow valleys.
- Avoid undoing previous progress.
- Goal:
- Move efficiently toward the minimum.

Conjugate Directions

- Two directions are **conjugate** if:
- Moving along one does not ruin the minimization along the other.
- If we find N conjugate directions:
- A quadratic function can be minimized in N line searches.
- Key idea:
- Directions should not interfere with each other.

Powell's Method

- Start with coordinate directions.
- Run linmin along each direction.
- Measure the net movement direction.
- Add this new direction.
- Remove one old direction.
- Repeat until improvement is small.



Overall Summary

- Why powell works better:
- Learns good directions automatically
- Avoids zig-zagging
- Faster convergence than coordinate descent
- Works without gradients

10.6: How to move optimally along one direction (linmin). 10.7: How to choose and update directions intelligently (Powell). Together they solve multidimensional minimization efficiently.